Class 018 - Mini Project

Ricardo Carias (PID: A18573289)

Table of contents

Background 1
Focus on Pretussis Toxin (PT) 16

Background

Pertussis (Whooping Cough) is a common lung infection caused by the bacteria B. pertussis. Most incompetent individuals will clear the infection. However, this infection can be deadly in infants.

As the Center for Disease Control (CDC) tracks the number of reported cases int he US:

Link: https://www.cdc.gov/pertussis/php/surveillance/pertussis-cases-by-year.html?CDC_AAref_Val=https://w
tussis/surv-reporting/cases-by-year.html

We can 'scrape' this data with the **datapasta** package.

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
    1931L,
    1932L,
    1933L,
```

```
    1934L,
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
    1948L,
    1949L,
    1950L,
    1951L,
    1952L,
    1953L,
    1954L,
    1955L,
    1956L,
    1957L,
    1958L,
    1959L,
    1960L,
    1961L,
    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
    1973L,
    1974L,
    1975L,
```

```
    1976L,
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
    1992L,
    1993L,
    1994L,
    1995L,
    1996L,
    1997L,
    1998L,
    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
    2015L,
    2016L,
    2017L,
```

```r
    2018L,
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
    68687,
    45030,
    37129,
```

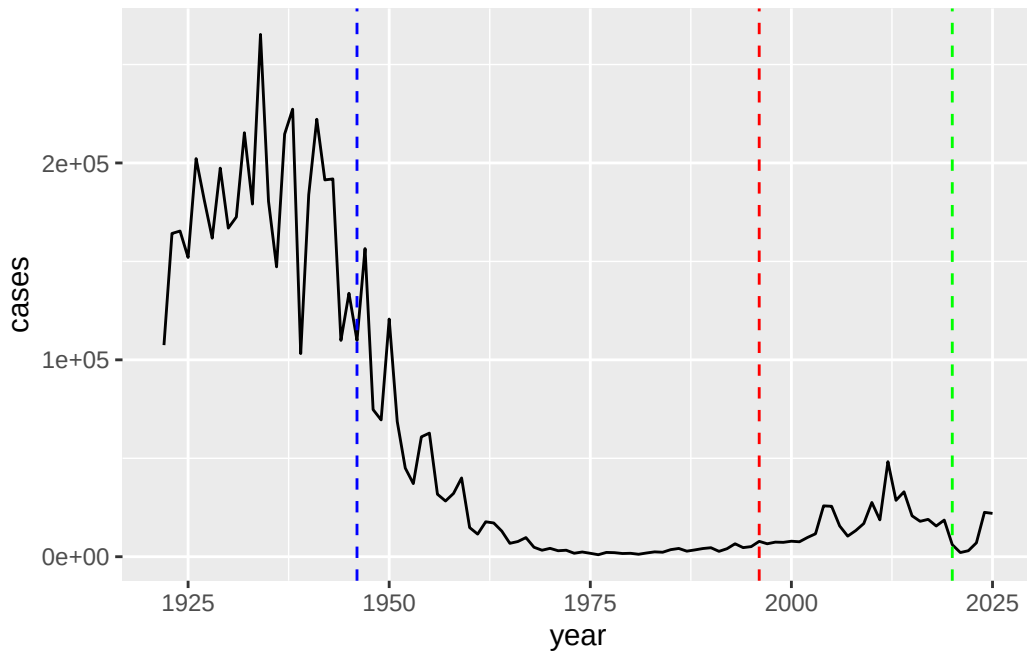
```
    60886,
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
    6586,
    4617,
    5137,
```

```
    7796,
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22538,
    21996

  )
)
```

Q. can we make a plot of these data?

```
library(ggplot2)

ggplot(cdc, aes(x = year, y = cases)) +
  geom_line() +
  geom_vline(xintercept = 1946, lty = 2, col = "blue") +
```

```
  geom_vline(xintercept = 1996, lty = 2, col = "red") +
  geom_vline(xintercept = 2020, lty = 2, col = "green")
```



> Q. Add some major milestones including the first wP vaccine (1946), the aP vaccine(1996), and COVID pandemic start at (2020).

Blue: wP vaccine introduced.

Red: Public opinion of vaccine declines and we see susceptible populations infection rates rise. Notably, around the 1996 time point, we have the rise of public access to internet and spread of misinformation.

Green: Covid_19 Pandemic

aP induced immunity weans faster than wP immunity. It appears that immunizations which stimulate a more robust immune response, due to the full set of inactivated components present, result in stronger immunological memory.

##CMI-PB Project (Computational Models of Immunity - Pertussis Boost)

This project aims to provide the scientific community with this very information.

Link: https://www.cmi-pb.org/

They make their data available via JSON format returning API. We an read this in R with the `read_jason()` from the **jsonlite** package:

```r
library(jsonlite)

subject <- read_json("http://cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many wP and aP individuals are in this `subject` table?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. Break down of biological sex?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. In terms of race and biological sex, is this dataset representative of US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more database tables:

```r
library(jsonlite)

specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)

ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)

head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we first need to merge these data frames / tables.

We can the `*_join()` family of functions from **dplyr** to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(meta, ab_titer)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
```

```
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           1
3    1986-01-01    2016-09-12 2020_dataset           1
4    1986-01-01    2016-09-12 2020_dataset           1
5    1986-01-01    2016-09-12 2020_dataset           1
6    1986-01-01    2016-09-12 2020_dataset           1
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit isotype is_antigen_specific antigen        MFI MFI_normalised  unit
1     1     IgE               FALSE   Total 1110.21154       2.493425 UG/ML
2     1     IgE               FALSE   Total 2708.91616       2.493425 IU/ML
3     1     IgG                TRUE      PT   68.56614       3.736992 IU/ML
4     1     IgG                TRUE     PRN  332.12718       2.602350 IU/ML
5     1     IgG                TRUE     FHA 1887.12263      34.050956 IU/ML
6     1     IgE                TRUE     ACT    0.10000       1.000000 IU/ML
  lower_limit_of_detection
1                 2.096133
2                29.170000
3                 0.530000
4                 6.205949
5                 4.679535
6                 2.816431
```

Q. What antibody isotypes are measured?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4
 6698  7265 11993 12000 12000 12000
```

Q. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
    PD1     PRN      PT     PTM   Total      TT
   1970    6712    6712    1970     788    6318
```

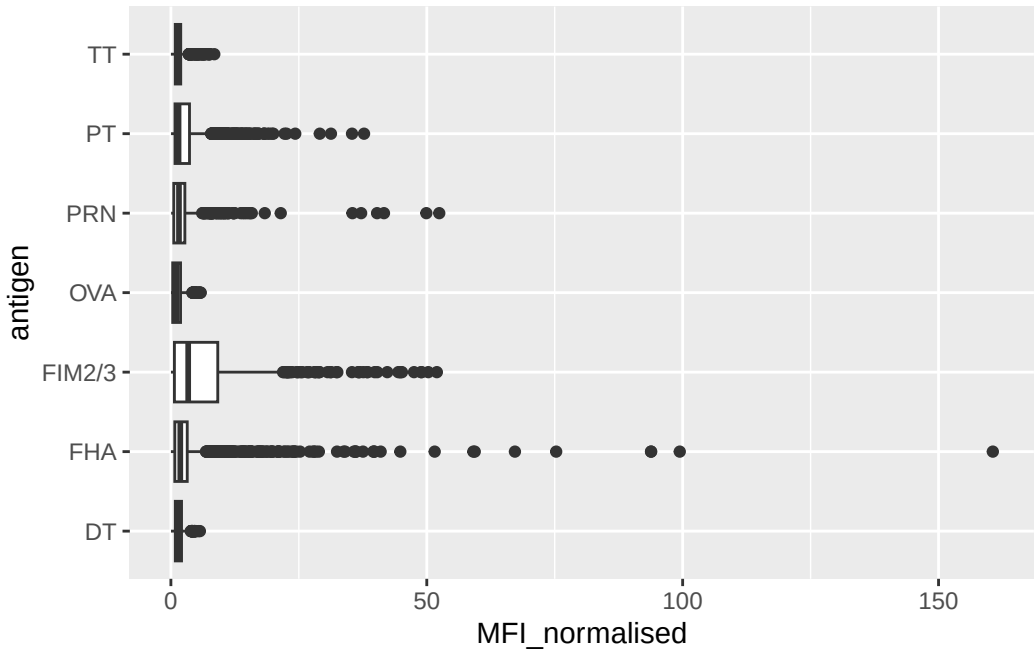
Focusing on IgG isotype and make a plot of MFI_normalized for all antigens.

```r
library(dplyr)

igg <- abdata|>
  filter(isotype == "IgG")

head(igg)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           1
3    1986-01-01    2016-09-12 2020_dataset           1
4    1986-01-01    2016-09-12 2020_dataset           2
5    1986-01-01    2016-09-12 2020_dataset           2
6    1986-01-01    2016-09-12 2020_dataset           2
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                            1                             1         Blood
5                            1                             1         Blood
6                            1                             1         Blood
  visit isotype is_antigen_specific antigen        MFI MFI_normalised  unit
1     1     IgG                TRUE      PT   68.56614       3.736992 IU/ML
2     1     IgG                TRUE     PRN  332.12718       2.602350 IU/ML
3     1     IgG                TRUE     FHA 1887.12263      34.050956 IU/ML
4     2     IgG                TRUE      PT   41.38442       2.255534 IU/ML
5     2     IgG                TRUE     PRN  174.89761       1.370393 IU/ML
```

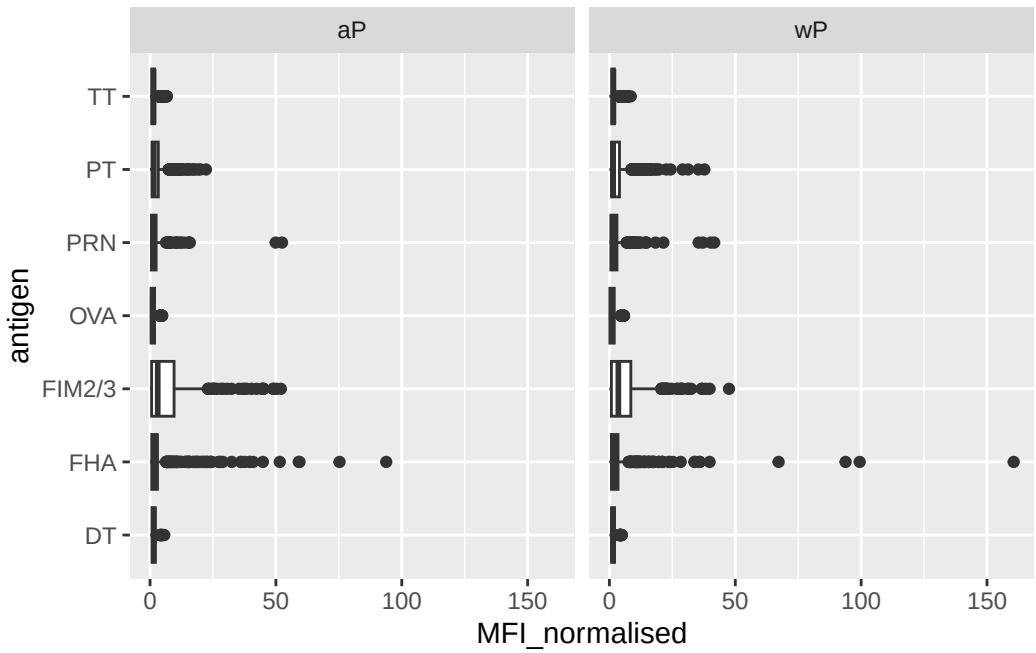
```
6     2     IgG                TRUE     FHA  246.00957       4.438960 IU/ML
  lower_limit_of_detection
1                 0.530000
2                 6.205949
3                 4.679535
4                 0.530000
5                 6.205949
6                 4.679535
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

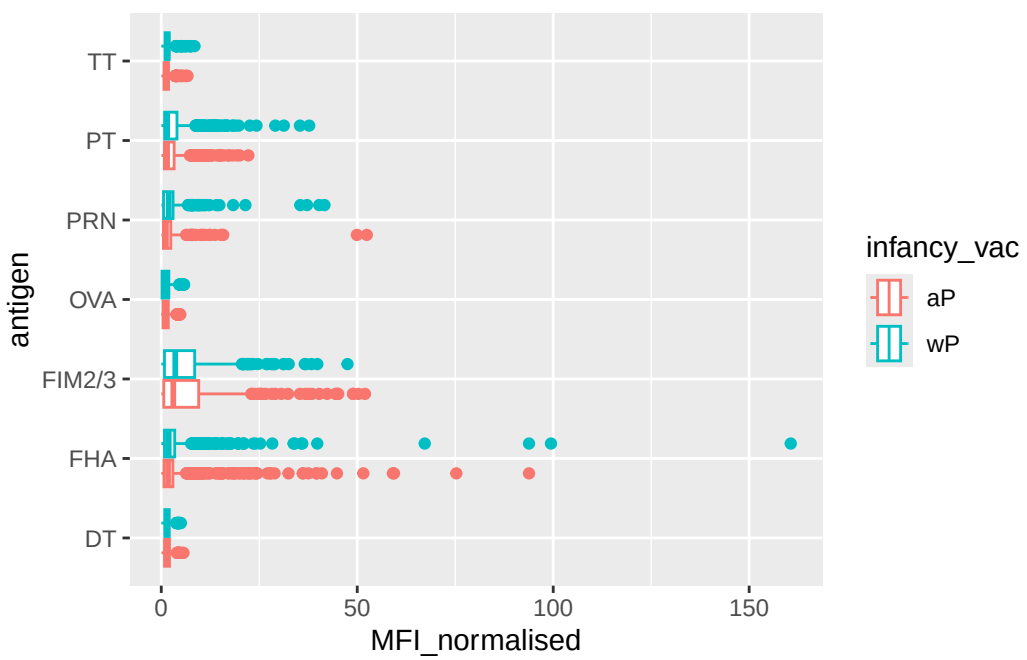


Q. Is there a difference for aP vs wP with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

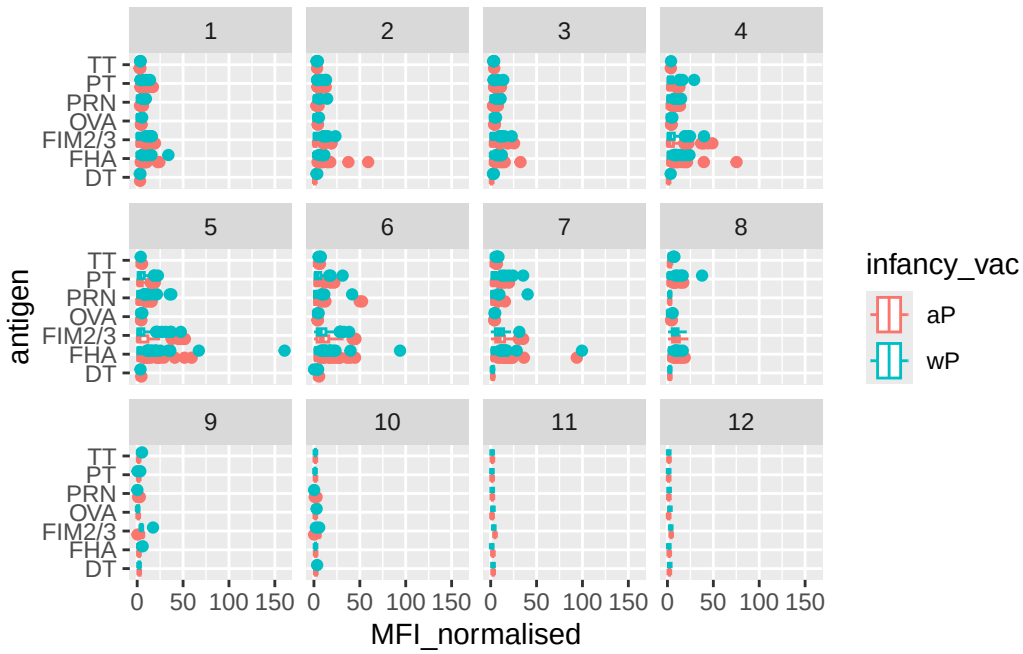



```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot()
```



Q. Is there a temporal response?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```



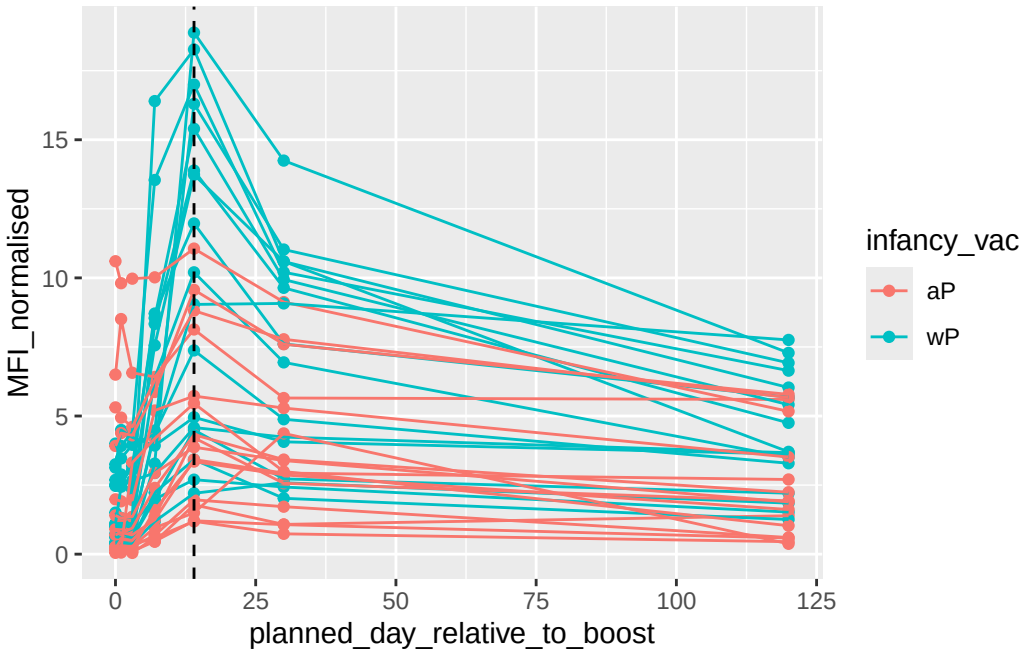
Focus on Pretussis Toxin (PT)

```
pt.igg.21 <- igg |> filter (antigen =="PT",
                dataset== ("2021_dataset")
                )
head(pt.igg.21)
```

```
  subject_id infancy_vac biological_sex              ethnicity
1         61          wP         Female Not Hispanic or Latino
2         61          wP         Female Not Hispanic or Latino
3         61          wP         Female Not Hispanic or Latino
4         61          wP         Female Not Hispanic or Latino
5         61          wP         Female Not Hispanic or Latino
```

```
6         61          wP         Female Not Hispanic or Latino
                     race year_of_birth date_of_boost      dataset specimen_id
1 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         468
2 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         469
3 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         470
4 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         471
5 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         472
6 Unknown or Not Reported    1987-01-01    2019-04-08 2021_dataset         473
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -4                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           14                            14         Blood
6                           30                            30         Blood
  visit isotype is_antigen_specific antigen    MFI MFI_normalised unit
1     1     IgG               FALSE      PT 112.75      1.0000000  MFI
2     2     IgG               FALSE      PT 111.25      0.9866962  MFI
3     3     IgG               FALSE      PT 125.50      1.1130820  MFI
4     4     IgG               FALSE      PT 224.25      1.9889135  MFI
5     5     IgG               FALSE      PT 304.00      2.6962306  MFI
6     6     IgG               FALSE      PT 274.00      2.4301552  MFI
  lower_limit_of_detection
1                 5.197441
2                 5.197441
3                 5.197441
4                 5.197441
5                 5.197441
6                 5.197441
```

```r
ggplot(pt.igg.21) +
  aes(x=planned_day_relative_to_boost,y= MFI_normalised, col=infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, lty=2 , col= "black")
```

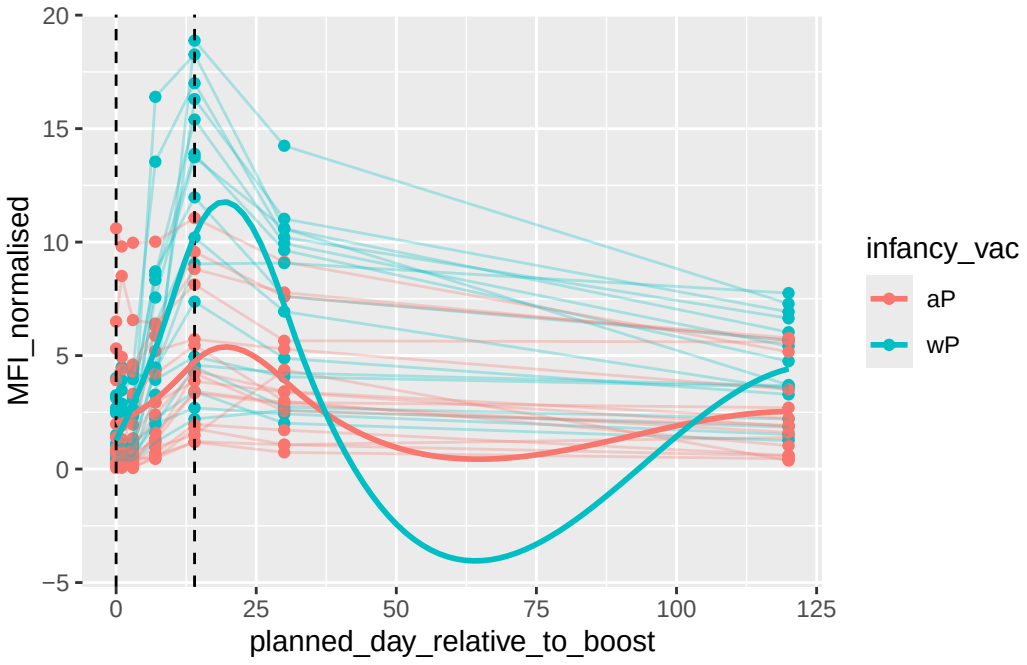


In the lab sheet there is one plot to try at home :

its actually in the slides actually.

```
ggplot(pt.igg.21) +
  aes(x = planned_day_relative_to_boost,
      y = MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.3) +
  geom_vline(xintercept = 14, lty = 2, col = "black") +
  geom_vline(xintercept = 0, lty = 2, col = "black") +
  geom_smooth(aes(group = infancy_vac), se = FALSE)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


20
15
10
5
0
−5
MFI_normalised
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP